Multimodal Federated Tea Disease Diagnosis

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***Abstract*—Tea pest and disease diagnosis benefits from lesion imagery and field notes, while federated adaptation can keep both records on-site. Small agricultural corpora, however, make multimodal and privacy claims easy to overstate. Our proposed model is a leakage-controlled five-class vision-language system with source-grouped splitting, exact image-box/text linkage, missing-modality tests, reliability-aware fusion, and measured FedAvg adaptation. The corpus contains 371 crops from 200 photographs, split into 222/74/75 train/validation/test crops from 122/40/38 disjoint sources; all 371 text records link to the exact box, and a training-fitted filter masks 178 target-shortcut tokens. On the identical 75-crop internal support under deterministic 50%-retention notes, frozen DistilBERT text-only, frozen ViT-tiny image-only, the raw ResNet–Transformer VLM, a new frozen ViT-tiny–DistilBERT VLM, and the proposed router obtain 65.33%, 66.67%, 77.33%, 82.67%, and 81.33% accuracy. Architecture selection is validation-only: the candidate's 82.43% validation accuracy remains below the incumbent's 89.19%, so ResNet-50 and the compact text Transformer are retained before candidate test inference. The candidate's one-crop internal-test edge is non-significant (exact McNemar $p = 1.0$), although its macro F1 is higher (0.8192 versus 0.6679). A separate same-checkpoint diagnostic reaches 92.00% with paired input but 33.33% after class-mismatching the text, demonstrating alignment sensitivity; its compact Transformer and ResNet paths are not relabelled as independent PLM and ViT baselines. In validation-only federated adaptation, final macro F1 over $K \in \{2, 3, 5, 8\}$ clients ranges from 0.8924 to 0.9099 versus a 0.9099 centralized initialization. Client identities are simulated rather than real estates, the test was observed during iterative development, and multi-site validation is required before privacy, deployment, or category-best claims.**

***Index Terms*—Federated Learning, Vision-Language Models, Tea Pest and Disease Classification, Multimodal Learning, Evaluation Protocol, Data Leakage, Precision Agriculture**

I. Introduction

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five visually confusable stress classes — leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug — reduce yield and leaf quality across major growing regions [1], [2]. Recent diagnostic systems improve field deployment through lightweight CNNs, attention modules, and edge-oriented pipelines [1]–[3], [5], [6], [8], but nearly all begin and end with the photograph. That is not how tea diagnosis is recorded on the ground, where staff note lesion shape, flush condition, weather history, and pest pressure alongside the image. Meanwhile, sending every high-resolution image and field note to a central repository is unattractive for estates with limited connectivity and proprietary records. Federated Learning [15] addresses the second problem, but existing FL work on tea pathology has not integrated the visual and textual evidence used in real inspections.

A second, methodological motivation became dominant during this study. One source photograph may contain several annotated regions, so crop-level random splits place near-identical leaf context on both sides of the boundary. Text can leak the target even more directly when disease names or class-pure terms remain in descriptions. We therefore group by source image, link text at the exact box level, and learn the blocked vocabulary from training annotations only. This changes the scientific question from "can a model memorise a disease description?" to "does each modality contribute on unseen source leaves?"

Contributions.

- We link every crop to one annotation by its (photograph, box index) key and keep source-image groups mutually exclusive across train, validation, and test.
- We fit a target-pure vocabulary on training text only, then evaluate paired, text-only, image-only, and mismatched-text conditions on the same 75 examples in the same label order.
- We compare the incumbent ResNet–Transformer VLM with a genuine ViT-tiny–DistilBERT VLM on the same 222/74/75 split. Both validation accuracy (89.19% versus 82.43%) and macro F1 retain the incumbent before candidate test embeddings and predictions are computed.
- We add a reliability-aware sparse-note rule that gains three correct crops and four accuracy points over the raw VLM, we keep its lower macro F1 in the report, and we identify the optional five-expert visual gate separately from a pure ViT baseline.
- We measure federated adaptation rather than assume it: a three-seed, validation-only FedAvg sweep over 2–8 label-skew clients and eight rounds retains 98.1–100.0% of the initialization's macro F1, and we preserve the centralized locked-test ablation while labelling simulated client ownership, internal routing, and missing multi-estate evidence explicitly.

II. Related Work

Tea disease detection. Tea-disease research progressed from standard backbone comparisons toward lightweight, attention-guided models. Deng and Photong compared VGG16, ResNet50, and DenseNet169 [3]. Yang et al. then combined wavelet denoising, MobileNetV3 features, channel–spatial attention, and focal loss in WaveLiteNet, and reported 98.70% accuracy with 3.16M parameters [1]. Hu et al. similarly combined segmentation with multiscale group attention and reported 97.43% [2]. Beyond tea, Joseph et al. constructed a real-time plant-disease dataset [5], Maurya et al. developed a lightweight meta-ensemble [6], and Shovon et al. combined multiple models in PlantDet [7]. Al-Shahari et al. extended diagnosis toward IoT-assisted crop management [8], while Shafik et al. synthesized the wider field [9]. This line remained image-only and centrally trained, and it did not report a source-photograph-disjoint audit comparable to ours.

Federated learning in agriculture. A parallel line of work moved training from a central server toward distributed agricultural devices. McMahan et al. established FedAvg as the basic aggregation method [15]. Later systems added different deployment properties: Tahir et al. incorporated explainability [10]; Devaraj et al. organized sensor and crop-health models hierarchically in RuralAI [11]; Ding et al. added blockchain coordination through Bc2FL [12]; Caminha and Oliveira used a cloud-driven design [13]; and Thonglek and Rattanatamrong evaluated decentralized edge learning [14].

TABLE I: Comparison with Related Work.

Work	Image	Text	FL	VLM	Leak. audit
Yang et al. [1], Hu et al. [2], Deng et al. [3]	✓	×	×	×	×
Joseph et al. [5], Maurya et al. [6]	✓	×	×	×	×
Tahir et al. [10], Devaraj et al. [11]	✓	×	✓	×	×
Ding et al. [12], Caminha et al. [13]	✓	×	✓	×	×
Wang et al. [27]	✓	✓	×	✓	×
Li et al. [28]	✓	✓	✓	✓	×
This work	✓	✓	✓	✓	✓

TABLE II: Source-grouped paired-data audit

Split	Sources	Crops	Blight	Hoppers	Rust	Looper/Mosq.
Train	122	222	77	5	40	61/39
Validation	40	74	26	2	14	20/12
Test	38	75	26	2	13	21/13
Total	200	371	129	9	67	102/64

These systems distributed image or sensor learning, but did not federate exactly paired image–text records.

Vision-language models. A third research direction connected visual evidence with language. Radford et al. aligned image and text embeddings through contrastive pre-training in CLIP [22]. Li et al. introduced Q-Former alignment in BLIP-2 [23], Alayrac et al. used gated cross-attention in Flamingo [24], and Yu et al. combined contrastive and captioning objectives in CoCa [25]. Agricultural work began to adopt this idea: Wang et al. applied a multimodal LLM to crop disease identification [27], while Li et al. combined frozen CLIP features with federated transfer learning in FedReplay [28]. The unresolved question was whether these multimodal benefits persisted on small, source-grouped, estate-scale data.

In Table I, this work was the only entry that reported a group-disjoint leakage audit beside its scores.

III. System Model

Figure 1 shows the deployment architecture. Each estate holds leaf photographs and agronomist annotations locally. The controlled test ablations use one centrally selected base checkpoint; a separate experiment then performs FedAvg adaptation of its trainable multimodal states while the visual backbone remains frozen. Text is tokenized by a deterministic 30 522-id hashed vocabulary at a maximum length of 128. Photographs are resized to 224×224 and ImageNet-normalized. The text branch produces token features and pooled $\mathbf{h}_t$, the vision branch produces a 7×7 spatial token map and pooled $\mathbf{h}_v$, and bidirectional attention forms the joint representation.

A. Dataset and Leakage Audit

The audited corpus contains 200 field photographs collected from the IIT Kharagpur tea garden and annotated with YOLO oriented bounding boxes. One crop is extracted per valid box with 10% context padding, giving 371 examples across *LEAF_BLIGHT* (129), *LEAF_HOPPERS* (9), *LEAF_RUST* (67), *LOOPER_CATERPILLARS* (102), and *MOSQUITO_BUG* (64). Classes are the raw YOLO identifiers under the supplied annotation schema.

Source isolation. Splitting is performed once with seed 42 over source filenames, not crops. Pairwise source overlap across the three partitions is zero, every crop appears exactly once, and the locked internal test therefore contains 75 crops from 38 unseen photographs. The nine-example leaf-hopper class is the main statistical bottleneck: only two examples occur in validation and two in test.

Exact pairing. The annotation table has one row per crop with a photograph identifier, a box index, a class id, a disease name, and text. Thus 371/371 records are linked by the exact (photograph, box index) key; train, validation, and test pairing coverage are all 100%. This corrects the label-aligned proxy pairing used in the earlier draft.

Text construction and leakage control. After selecting a class, the annotation workflow samples manually written symptom, observation, condition, and indicator phrases; some rows substitute cross-class distractors. It inserts them into a class-neutral sentence template and stores the result with the class id/name and exact photograph–box key. No LLM generates these rows, and they are templated annotations rather than independent farmer notes. From training text only, we mask tokens appearing at least three times with class purity at least 0.95; the resulting 178-token vocabulary is fixed for validation and test.

Image-data scope. All 371 exact image–text pairs used for fitting, model selection, and reporting come from the curated IIT Kharagpur tea-garden corpus, its YOLO box annotations, and the corresponding linked text records; no external tea-disease images are pooled. A separate proxy bundle is excluded from this diagnostic experiment. ImageNet contributes pretrained ResNet/ViT weights only, not additional tea samples.

B. Model Variants and Objective

Text branch. A 192-d token and positional embedding feeds two pre-norm Transformer layers (four heads, 576-d feed-forward block), followed by a 256-d projection and masked mean pooling. Hashing is stable across machines and does not require a mutable vocabulary file.

Vision branch. An ImageNet-pretrained ResNet-50 [4] is frozen through the selected training run. Its 2048-channel feature map is projected to 256 channels and adaptively pooled to 7×7, preserving 49 spatial tokens for lesion-level attention. The complete model has 32.20M parameters, of which 8.69M are trainable.

Cross-modal branch. One pooled summary from each modality queries the other modality through separate eight-head attention layers. A learned reliability network produces (r_t, r_v) with $r_t + r_v = 1$ over available modalities. The fusion head receives the weighted parent features, both cross-attended summaries, their absolute difference and element-wise product, and the modality-presence mask. Two linear expert heads then re-enter its logits $\mathbf{z}_f$ as a fixed reliability-weighted residual, $\mathbf{z} = \mathbf{z}_f + 2(r_t\mathbf{z}_t + r_v\mathbf{z}_v)$, each expert zeroed when its modality is absent. The residual is active at test time, so every reported multimodal score includes it.

Independent comparison models. The requested family comparison does not alter or rename these core branches. Its text-only pretrained language model (PLM) is a frozen `distilbert-base-uncased` encoder [33] with masked-mean embeddings and a validation-selected class-weighted ridge head. Its image-only model is a frozen `vit-tiny-patch16-224` encoder [17] with a validation-selected ridge head. The image+text VLM is the unchanged raw bidirectional cross-attention output, before any post-hoc class gate; the proposed row is that same core plus the existing reliability-aware router. A direct replacement candidate combines the frozen ViT and DistilBERT encoders, train-fitted 64-component projections, and a class-weighted linear fusion head. Its 360 configurations use validation only; test embeddings are extracted after freezing the choice. DistilBERT is therefore described precisely as a PLM, not as an instruction-tuned generative LLM. The proposed architecture remains

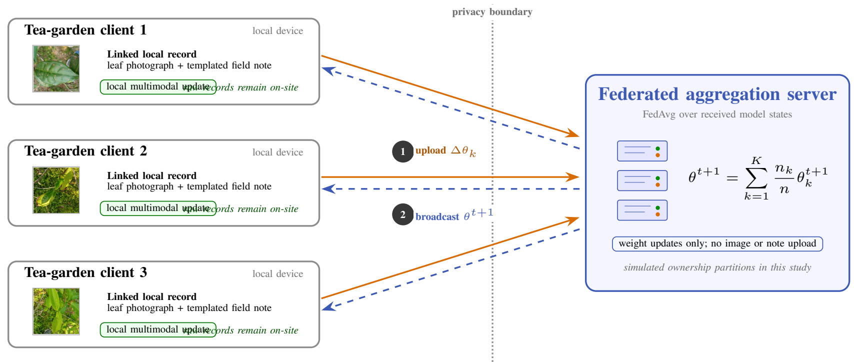


Fig. 1: Federated deployment workflow using representative corpus photographs. Each simulated tea-garden client keeps linked photographs and field notes on-site; only model updates $\Delta\theta_k$ cross the explicit privacy boundary, and the FedAvg server returns θ^{t+1}.

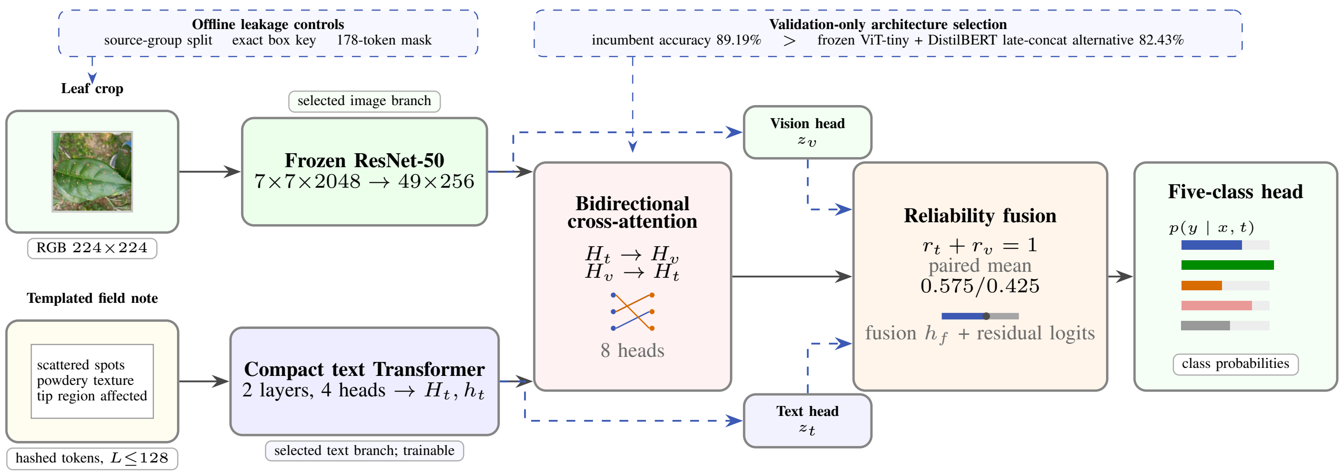


Fig. 2: Active inference architecture with a real corpus crop and a representative templated note. The dashed blue boxes separate offline leakage controls and validation-only system selection from runtime inference. The selected ResNet–compact-Transformer core uses bidirectional cross-attention; dashed runtime paths are explicit unimodal expert heads that re-enter the fusion logits. The pooled ViT–DistilBERT alternative is screened separately and is not drawn as token-level cross-attention.

the ResNet-50, compact text Transformer, cross-attention, and router shown in Figs. 2 and 3.

For one-hot label $\mathbf{y}$, training uses class-weighted, label-smoothed cross entropy with auxiliary parent losses and supervised cross-modal alignment:

$$\mathcal{L} = \mathcal{L}_f + 0.20\mathcal{L}_t + 1.25\mathcal{L}_v + 0.05\mathcal{L}_{align}. \tag{1}$$

The visual weight is deliberately larger because image-only learning is the harder path. Modality dropout samples image-only, text-only, and paired batches with probabilities 0.50, 0.10, and 0.40. Checkpoint selection uses $0.65F_{1,\text{paired}} + 0.35F_{1,\text{image}}$ on validation.

For K clients, the same local objective is aggregated with FedAvg [15],

$$\theta^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \theta_k^{t+1}, \tag{2}$$

where n_k is client k's sample count. Our federated experiment initializes from the selected base state, freezes ResNet-50, and averages every remaining multimodal parameter and floating buffer after one local epoch. This is federated *adaptation*, not training from scratch; its validation curves do not relabel the centralized 75-crop test result.

TABLE III: Implemented model blueprint

Block	Specification	Output / training state
Text encoder	192-d embedding; 2 Transformer layers; 4 heads	256-d pooled vector; trainable
Vision encoder	ImageNet ResNet-50; 224^2 input	49×256 tokens; frozen backbone
Cross-attention	Two directional layers; 8 heads	text-to-image and image-to-text summaries
Reliability fusion	normalized (r_t, r_v) plus interaction features	5-way trainable classifier
Expert residual	linear text and vision heads; fixed $\lambda = 2$	added to the fusion logits at train and test
Complete model	32.20M parameters	8.69M trainable parameters

IV. Proposed Framework

A. Base Cross-Modal Checkpoint

The base checkpoint trains for 15 epochs with AdamW (batch 16, peak learning rate 10^{-4}, weight decay 10^{-4}, two-step gradient accumulation). Two independently augmented frozen-ResNet feature views (crop/orientation/colour perturbations) are cached per training crop. Hence $222 \times 2 = 444$ cached training views represent 222 unique crops, not 444 independently collected leaves; the complete corpus remains 371 unique crops. Validation and test remain 74 and 75 crops. The six-view test-time expert below averages predictions for one crop and does not enlarge the test set. The first four epochs emphasize image-only warm-up, while the confidence override used in an earlier draft is disabled. The selected epoch is fixed

TABLE IV: Class-wise visual experts selected on validation

Base class	Selected expert	Test recall
Leaf blight	Deep feature	0.692
Leaf hoppers	Calibrated base	0.500
Leaf rust	Frozen ViT	0.615
Looper caterpillars	Base TTA	0.714
Mosquito bug	Frozen ViT	0.923

before the 75-crop test is evaluated, and the checkpoint reloads with strict key matching.

B. Enhanced Image-Only Inference

The direct image path is the main weakness, so we retain the base checkpoint but construct an optional visual inference system from five independent experts:

1) direct base image-only prediction;
2) six-view deterministic test-time augmentation;
3) a frozen ResNet deep-feature specialist;
4) validation-fitted class calibration; and
5) a frozen ImageNet-pretrained ViT-tiny specialist.

For base predicted class c, a gate g_c selects one expert. All $5^5 = 3125$ gates are scored on the 74-crop validation partition using $0.6F_1 + 0.4\,\text{accuracy}$. The selected vector $\mathbf{g} = [2, 3, 4, 1, 4]$ is frozen before test inference. It raises validation accuracy from 66.22% for the direct image path to 85.14%; Section V reports the held-out result.

C. Reliability-Aware Sparse-Note Rule

To study incomplete field notes, each non-special text token is retained by a stable hash with probability 0.50; the same record always receives the same mask. The *late-fusion baseline* starts from the paired cross-attention prediction and uses the enhanced image expert for base-predicted leaf-hopper and mosquito-bug cases, the validation-selected gate $[0, 1, 0, 0, 1]$. The proposed rule is more conservative: when raw fusion confidence is below 0.60, only base-predicted leaf-rust cases route to the sparse-text expert; all other cases retain late fusion. This rule is observable, deterministic, and does not inspect test labels.

This experiment has a stricter limitation than the standard ablation. Candidate sparsity families and routing constraints were refined after observing the internal partition, so the final 75-crop result is **exploratory and non-pristine**. It demonstrates a mechanism and supplies an internal development metric; it is not unbiased external validation.

D. Multimodal Federated Adaptation

The federated track starts from the validation-selected base checkpoint, keeps the 23.51M-parameter ResNet-50 encoder fixed, and adapts the remaining 8.69M multimodal parameters. Training crops are partitioned per class with a Dirichlet draw ($\alpha = 1$); full participation and sample-count weighting are used for one local epoch per round. We sweep $K = \{2, 3, 5, 8\}$ for eight rounds over client-partition seeds 42, 123, and 2026. Every curve is evaluated on the 74-crop, 40-source validation partition; the 75-crop locked test is not queried.

The clients are simulated ownership partitions because the corpus has no estate identifier. The experiment measures optimization stability of multimodal FedAvg adaptation, not real cross-estate privacy, communication security, or generalization. The visual ensemble and sparse-note router remain centralized inference layers and are not silently included in the federated score.

TABLE V: Validation-only multimodal FedAvg adaptation. Values are means over three client-partition seeds.

K	Final F1	Best F1	Retained	Label TV
2	0.8924	0.9099	98.1%	0.139
3	0.9046	0.9237	99.4%	0.217
5	0.9099	0.9297	100.0%	0.270
8	0.9099	0.9099	100.0%	0.300

E. Advisory Module

The framework includes a Classify–Retrieve–Advise module: a query is encoded by Sentence-BERT [35], FAISS [34] returns top-k passages from a per-estate treatment base, fusion features re-rank them, and the top passage becomes the recommendation [36]. A companion visualizer overlays class-coloured OBB polygons on uploaded or retrieved reference photographs.

Earlier drafts left this module unevaluated because that run's corpus was empty. We now supply one from the bundled symptom-text collection, which carries the same five class identifiers but is a *different* corpus from the 371 crop-linked notes used for diagnosis: it is templated, unlinked to any photograph, and evaluated on its own support, so no number here is comparable to the 75-crop test and the two are never pooled. Its 3,000 annotations collapse to 673 distinct observations built from 131 sentences, 43 of which appear under more than one class. After deduplication we hold out 30% of each class, leaving a 472-passage base and 201 queries sharing no observation. Queries are encoded with `all-MiniLM-L6-v2` and scored by exact inner product over unit vectors, which is what a FAISS `IndexFlatIP` computes, so the index adds no approximation. A passage is correct only if it carries the query's class, since a wrong-class passage is wrong treatment advice.

V. Performance Evaluation

A. Protocol

The base checkpoint was trained in PyTorch [37] for at most 15 epochs. All crops from a source photograph remain in one partition. Model selection and the five-expert visual gate use only the 74-crop validation partition; every number below uses the same 75-crop, 38-source internal test support. Accuracy is accompanied by macro F1 because the class distribution is imbalanced. Source-group bootstrap intervals (5,000 replicates) resample source photographs rather than treating multiple crops from one photograph as independent observations.

Four evidence tracks are kept separate. (i) The *standard same-checkpoint ablation* uses the compact in-model Transformer and ResNet paths with original paired descriptions; these paths are not independent PLM or ViT baselines. (ii) The *current family comparison* evaluates actual frozen DistilBERT, frozen ViT-tiny, raw cross-attention VLM, the new ViT–DistilBERT VLM, and unchanged proposed predictions on identical validation and test label sequences under the deterministic 50%-retention note mask. The replacement configuration is frozen on validation before test embedding; the comparison remains exploratory and non-pristine. (iii) The *federated sweep* varies clients and rounds on validation only. (iv) The *earlier within-family screen* compares candidates only inside its text, vision, or proxy-fusion panel because those panels use 15 template groups, 30 source images, and 30 unlinked label-aligned pairs, respectively.

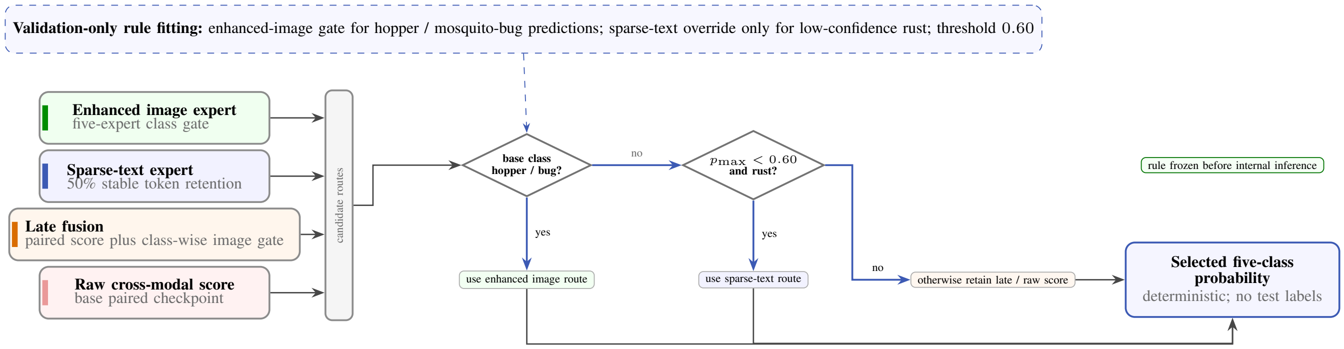
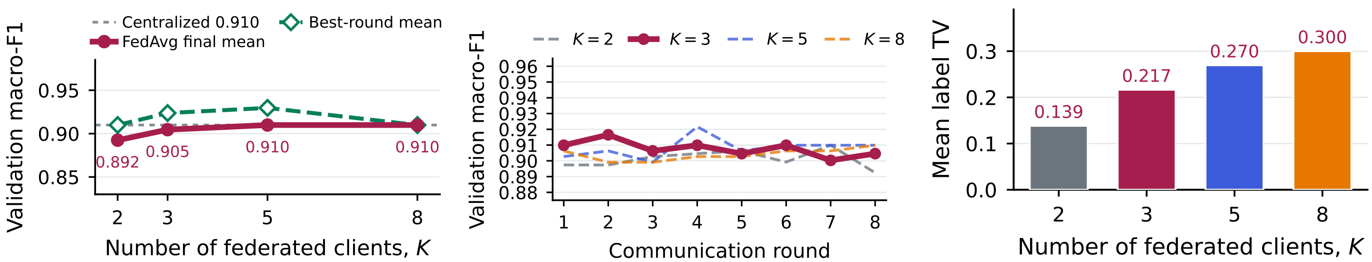


Fig. 3: Reliability-aware sparse-note routing as an explicit decision flow. Four available prediction routes feed a deterministic class/confidence rule selected on validation. Blue arrows show the frozen decisions; the router does not read test labels, and the reported internal result remains exploratory and non-pristine.



(a) Client count versus final and best-round performance.

(b) Communication rounds; the deployed $K = 3$ trace is emphasized.

(c) Realized label skew actually induced at each client count.

Fig. 4: Measured multimodal FedAvg adaptation on the source-disjoint tea validation split. Points are mean macro F1 over three client-partition seeds; 222 training crops, 74 validation crops, $\alpha = 1$, one local epoch, full participation, and a frozen ResNet-50 are fixed. The dotted line in (a) is the 0.9099 initialization, and (c) is the realized label total variation, so (a) is read against measured rather than assumed skew. The 75-crop test is not used, and client ownership is simulated rather than estate-derived.

TABLE VI: Standard same-checkpoint ablation. These are internal core paths, not the independent PLM/ViT/VLM systems in Fig. 5.

Condition	Accuracy	Correct	Macro F1	ECE
Core image path (ResNet)	0.6133	46/75	0.5431	0.1135
Core text path (compact Tr.)	**0.9600**	72/75	**0.9049**	0.1463
Base paired VLM	0.9200	69/75	0.8753	0.2648
Mismatched-pair VLM	0.3333	25/75	0.2804	0.2193

B. Federated Client and Round Scaling

Figure 4 and Table V answer the two federated scaling questions directly. Final-round macro F1 ranges from 0.8924 to 0.9099 and is never perfect. At the predeclared $K = 3$ setting, the final mean is 0.9046, or 99.4% of the 0.9099 initialization. Round-wise means stay between 0.8924 and 0.9216; the highest transient mean occurs for $K = 5$ at round 4, but it is reported as a validation peak rather than the final model. Increasing K raises realized label-distribution total variation from 0.139 to 0.300 without a monotone final-score penalty.

This stability result is useful but narrow. Every run begins from the same centrally selected checkpoint, so it supports federated adaptation of the multimodal core, not superiority over training from scratch. Three partition seeds measure ownership-split variability, not full checkpoint-training variance, and simulated clients cannot establish estate-level privacy.

C. Standard Same-Checkpoint Ablation

Table VI is the primary controlled comparison. The paired model is realistic rather than perfect: it predicts 69 of 75 crops correctly. Text only is higher at 72 of 75, so this experiment does *not* support a claim that standard fusion beats its strongest parent. Replacing each image's description with a description from another class drops accuracy from 92.00% to 33.33% and macro F1 from 0.8753 to 0.2804. The checkpoint therefore uses cross-modal association, but not in a way that improves over the unusually informative annotation-derived text. The independent family comparison instead applies the common sparse-note condition and actual DistilBERT/ViT implementations, so its lower scores answer a different, harder question and are not contradictory.

D. Raising Image-Only Accuracy

The optional five-expert visual gate combines the direct base path, deterministic test-time augmentation, a deep-feature specialist, calibration, and ViT-tiny. Validation-only selection over 3,125 class gates raises image accuracy from 61.33% (46/75) to **72.00%** (54/75), with macro F1 0.6759 and source-group bootstrap intervals 60.23–85.14% and 0.5365–0.8256. This is an ensemble result, not the pure ViT baseline (66.67%). Its gains concentrate in leaf blight and looper caterpillars, while only two leaf-hopper crops make rare-class behavior unstable.

E. Actual Model-Family and Inter-Model Comparison

DistilBERT is frozen (66.36M parameters); validation selects masked-mean L2 features, ridge $\alpha = 0.1$, and class-weight power 1 from 90 candidates before test embeddings are extracted. ViT-tiny is independently pretrained and frozen, with its CLS-feature ridge head also selected on validation. The raw VLM row is the unchanged paired bidirectional cross-attention prediction under the same 50%-retention note

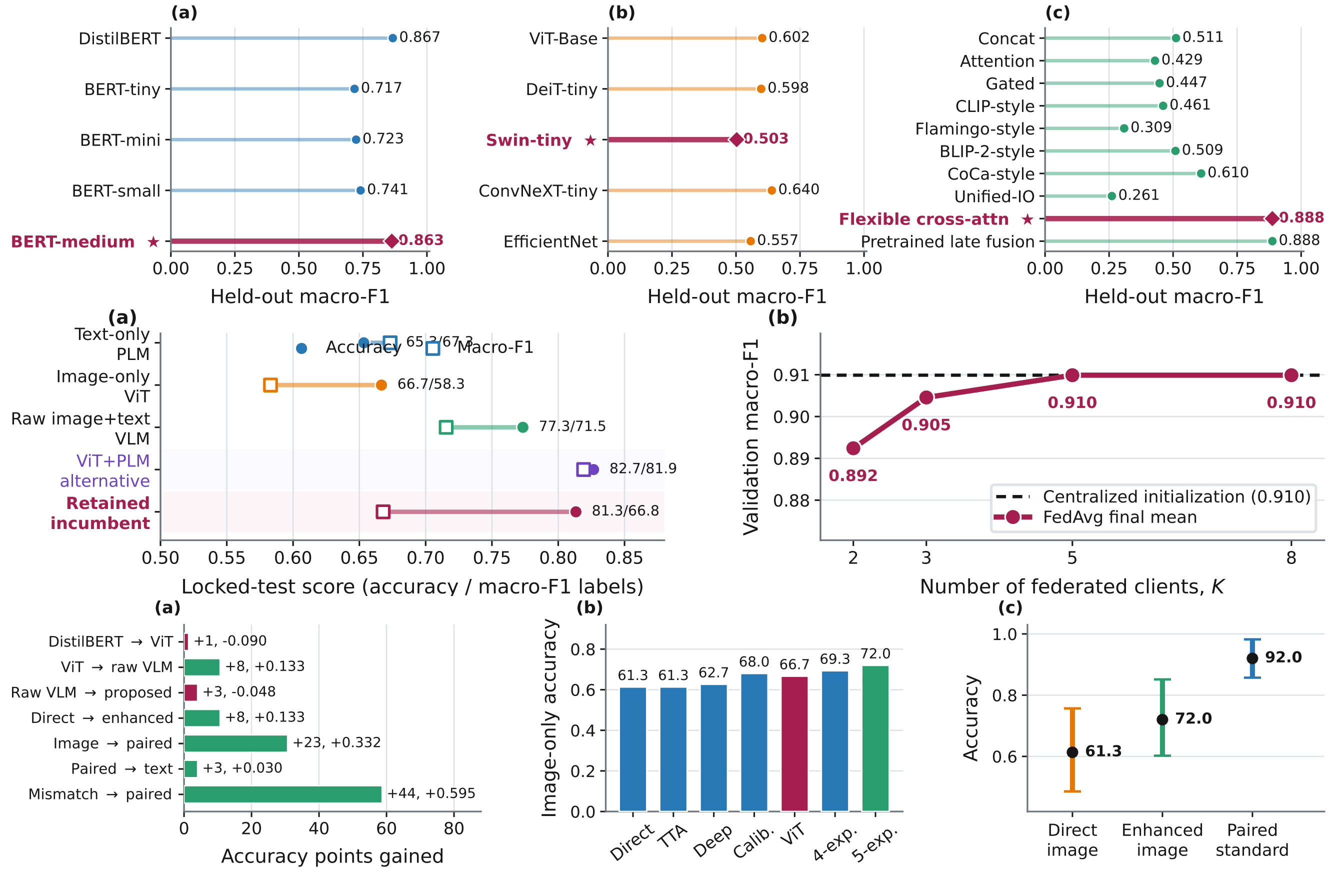


Fig. 5: Model comparisons without changing the proposed core. *Upper:* earlier within-family screening by held-out macro F1: (a) five text PLMs on 15 template groups, (b) five visual backbones on 30 source-disjoint images, and (c) ten fusion-rule surrogates on 30 label-aligned proxy pairs. Stars mark the validation-selected representative; supports differ across panels, so no cross-panel ranking is valid. *Middle:* (a) actual DistilBERT, ViT-tiny, raw cross-attention VLM, pooled ViT–DistilBERT alternative, and unchanged proposed model on the same ordered 75 crops under sparse notes; (b) centralized proposed initialization versus final multimodal FedAvg adaptation on 74 validation crops. *Lower:* fixed 75-crop decompositions — (a) every pairwise contrast as accuracy points gained, annotated with the correct-crop change and Δmacro F1, where red bars mark the two contrasts that gain accuracy while losing macro F1; (b) the image-only ladder, in which each stage adds a component rather than retraining; and (c) source-group bootstrap accuracy over 5,000 replicates, whose two image intervals overlap. Accuracy is in percent labels while the plotted scale is 0–1. Plot titles are omitted to maximize legibility.

TABLE VII: Common-support encoder selection. Architecture choice uses validation only; test metrics are reported after selection.

Actual system	Val. acc.	Test acc.	Test F1
Text PLM (DistilBERT)	0.6757	0.6533	0.6732
Image ViT (ViT-tiny)	0.7973	0.6667	0.5830
Raw ResNet+compact VLM	0.8378	0.7733	0.7155
ViT+DistilBERT VLM	0.8243	**0.8267**	**0.8192**
Proposed ResNet+compact	**0.8919**	0.8133	0.6679

mask and no post-hoc class gate. Ordered labels, predictions, support, and recomputed metrics are checked to be identical across all rows. The replacement VLM uses two reproducible augmented training views, frozen ViT-tiny and DistilBERT embeddings, 64 train-fitted components per modality, and a validation-selected class-weighted ridge head ($\alpha = 100$, weight power 0.5). Among 360 predeclared configurations, it reaches 82.43% validation accuracy and 0.6892 macro F1, below the incumbent's 89.19% and 0.8541. The ResNet–compact-Transformer architecture is therefore retained before candidate test features and predictions are computed. This candidate is pooled late feature fusion; it does not use token–patch cross-attention.

Consequently, the comparable accuracy hierarchy is

$$65.33\%_{\mathrm{PLM}} < 66.67\%_{\mathrm{ViT}} < 77.33\%_{\mathrm{raw}}$$
$$< 81.33\%_{\mathrm{proposed}} < 82.67\%_{\mathrm{ViT+PLM}}.$$

The candidate later predicts one more internal-test crop correctly than the retained model, but this cannot retroactively change a validation-only choice. The paired discordance is 11 candidate-only versus 10 incumbent-only correct (exact two-sided McNemar $p = 1.0$). Macro F1 is also higher for the candidate, 0.8192 versus 0.6679. Figure 5 therefore separates the selection result from descriptive internal-test outcomes.

Within-family screens. The upper row of Fig. 5 supplies the requested model-level comparisons. In the earlier text screen, DistilBERT, BERT-tiny/mini/small, and validation-selected BERT-medium are compared on 15 held-out template groups. The visual screen compares ViT-Base, DeiT-tiny, Swin-tiny, ConvNeXT-tiny, and EfficientNet on 30 source images; Swin is the validation selection although ConvNeXT has the higher held-out macro F1. The fusion screen compares ten implemented fusion-rule surrogates on 30 label-aligned proxy pairs; "CLIP-", "Flamingo-", "BLIP-2-", and related labels indicate fusion styles rather than deployments of those pretrained systems. These three supports are not linked observations, so only within-panel comparisons are valid and they are not mixed with Table VII.

F. Cross-Modal Analysis and Evidence Boundary

Under paired input the learned mean reliability weights are 0.5749 for text and 0.4251 for image. Mismatching text causes a 58.67-point accuracy drop and a 0.5949 macro-F1 drop, confirming that the cross-attention path is sensitive to alignment. Retrieval analysis is more restrained: exact text-to-image recall@1 is 1.33%, while class-level text-to-image and image-to-text recall@1 are 0.6000 and 0.6133. The model organizes class evidence better than exact crop identity, consistent with descriptions that encode disease labels more strongly than instance-specific appearance.

G. Architecture Component Ablation

Nine variants were each retrained from scratch for 15 epochs on the identical split and scored on the same locked crops, under two runs differing only in hardware (CPU/PyTorch 2.13 and H100/PyTorch 2.11). On the H100 the unchanged model is best at 94.67% (71/75) and 0.8957 macro F1, and every ablation costs accuracy: removing cross-attention or the auxiliary parent losses gives 81.33% (61/75) and removing interaction features 82.67% (62/75). That table is nonetheless not reportable as a component ranking. The same unchanged configuration scores 85.33% on the CPU run, a 9.34-point gap with no change of code, seed, or data; the mean absolute gap across variants is 4.00 points, and five of eight measured effects reverse sign between runs. Component differences here are therefore smaller than the run-to-run variance floor, and a stable ranking would need several seeds per variant rather than one.

H. Advisory Retrieval Results

Retrieval alone returns a correct-class passage for 90.05% of queries (95% CI 85.57–94.03) against a 20.01% chance rate, with precision@5 93.23%, MRR 0.9469, and NDCG@5 0.9621. The classify stage raises this to 95.52% (95% CI 92.54–98.01), but that second number *is* the classifier's accuracy: retrieval restricted to one predicted class can only return that class, so routing rather than ranking decides correctness. Fig. 6(b) locates the failure. On the 65.67% of queries carrying a class-unique sentence open retrieval is near-saturated at 99.24%; on the rest it falls to 72.46%, which routing lifts to 89.86%. Per class it is weakest on looper caterpillars (77.50%), whose vocabulary overlaps mosquito bug damage.

Three limits apply. The corpus is template-composed, so base and queries share sentence vocabulary even though no observation is shared; these are upper bounds relative to free-form notes. The fusion re-ranker needs a fusion model on these five classes and is still unevaluated, as are the OBB overlay and the agronomic correctness of the advice itself. This is a retrieval result, not evidence that the recommended treatment is right.

I. Reproducibility Manifest

The updated bundle records data identity, checkpoint state, every federated round, and each realized client partition. Table VIII separates mechanically verified controls from properties outside the evidence. Deterministic grouping, strict reload, and validation-only adaptation are reproducible; external estate transfer is not inferred from them.

TABLE VIII: Reproducibility manifest for the updated run

Control	Recorded value	Status
Source split	seed 42; 122/40/38 sources	fixed
Cross-split overlap	0 source photographs	verified
Crop–text pairing	371/371 exact box keys	verified
Shortcut vocabulary	178 tokens; train fitted	fixed
Checkpoint reload	strict key match	passed
Architecture ablation	9 variants; CPU/H100 reruns	single seed each
DistilBERT selection	90 candidates; validation only	fixed before test embed.
ViT+PLM VLM screen	360 candidates; validation accuracy	incumbent retained
Visual gate search	3,125 gates on validation	completed
Internal test support	75 crops / 38 sources	common
FedAvg adaptation	$K = 2, 3, 5, 8$; 8 rounds	completed
FL repetitions	seeds 42/123/2026	partition seeds
FL evaluation	74 validation crops	test untouched

J. Class-Level Error Accounting

Figure 7 resolves the 75 crops by class, with the diagonals summing to the 46, 54, and 61 correct crops of the three systems. The enhanced gate's net gain of eight is concentrated in leaf blight (+5), leaf hoppers (+1), and looper caterpillars (+4), partly offset by one fewer correct rust and mosquito-bug crop. Relative to the direct image model, the exploratory router adds 15 correct predictions overall: +7 blight, +1 rust, and +8 looper, offset by one mosquito-bug error and no leaf-hopper recovery. The off-diagonal cells locate the residual failure: leaf blight is confused with rust and looper rather than scattered, and the leaf-hopper crops fall into looper, one of which only the enhanced gate recovers. This accounting explains why aggregate accuracy rises while rare-class behavior remains unstable.

K. Uncertainty Facts

Source-group bootstrap interval widths are 27.11 points for the direct image model, 24.91 for the enhanced image gate, and 12.51 for standard paired input. The corresponding error counts are 29, 21, and 6 of 75; the exploratory router makes 14 errors. These intervals describe uncertainty under resampling of the 38 source photographs and do not replace a paired test of model differences. They also cannot quantify estate shift because every crop comes from the same audited corpus.

L. Threats to Validity

The internal test contains only 38 source photographs and 75 crops, with two leaf-hopper crops; bootstrap uncertainty and per-class instability are therefore substantial. The descriptions originate from the annotation workflow rather than independent farmer notes, explaining the very strong text baseline and limiting ecological validity. Although the visual gate is selected on validation data, choosing among five experts on 74 validation crops can still overfit. The central checkpoint has one training seed; the three FL seeds vary client partitions, not checkpoint initialization. Simulated clients lack estate ownership, attack, communication, and privacy accounting. Most importantly, iterative sparse-note and routing development makes that test non-pristine. DistilBERT is a pretrained language-model baseline rather than an instruction-tuned generative LLM; earlier fusion labels are implemented surrogates, and their differing supports forbid cross-panel ranking. A prospective multi-estate set, independent field notes, more rare-class cases, repeated end-to-end training, and a from-scratch federated comparison are required before deployment claims.

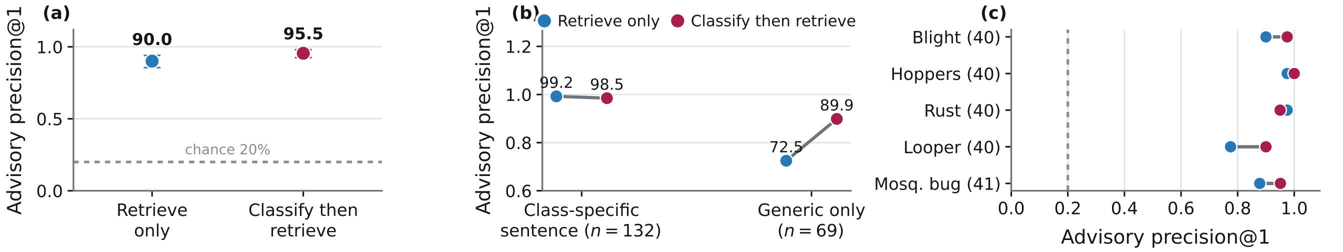


Fig. 6: Measured Classify–Retrieve–Advise quality on the templated symptom corpus: 472 passages, 201 queries, exact inner-product search. (a) Precision@1 with bootstrap intervals against chance; (b) split by whether the query holds a class-unique sentence; (c) per class. Same five classes as the diagnostic experiments, but a different corpus and support, never pooled with them.

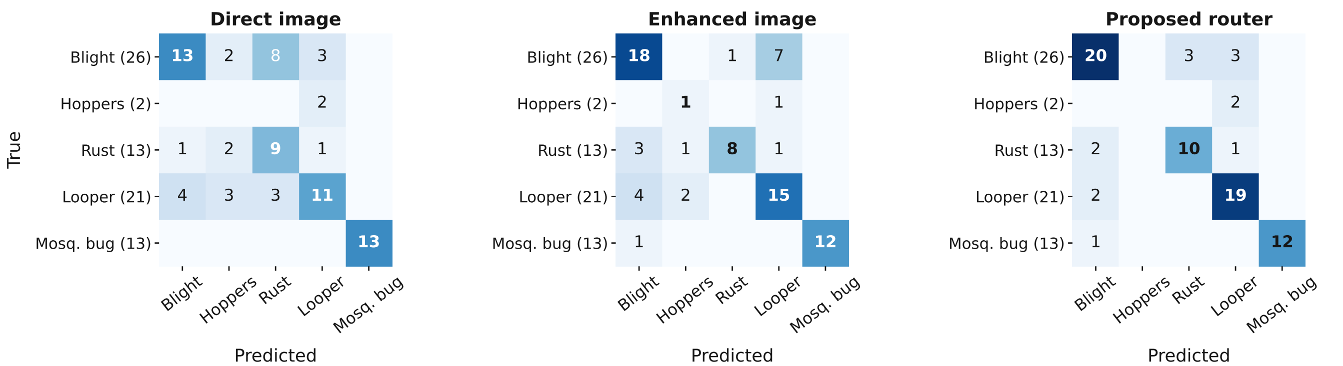


Fig. 7: Where the errors land on the fixed 75-crop test. Each matrix sums to 75; diagonals give the 46, 54, and 61 correct crops of the direct image path, the five-expert enhanced gate, and the exploratory router. All panels share one count colour scale, so shading is directly comparable. The off-diagonal cells are the part no count table records: leaf blight is confused with rust and looper rather than scattered, looper recovers most under the router, and the two leaf-hopper crops fall into looper except for one the enhanced gate recovers.

VI. Conclusion

With 371 linked crop–text pairs from 200 source photographs, validation-only selection retains the ResNet-50–compact-Transformer router over the frozen ViT-tiny–DistilBERT candidate. The candidate is one crop higher on the 75-crop test, but not significantly so ($p = 1.0$); paired CPU/H100 ablations also give no stable component ranking.

Warm-start FedAvg retains 98.1–100.0% of initial validation macro F1 for 2–8 simulated clients, and mismatched text confirms alignment sensitivity. These internal results do not establish category-best performance, privacy, or deployment readiness. Next steps are locked multi-estate evaluation, independent notes, rare-class coverage, secure aggregation, communication accounting, and repeated from-scratch training.

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